1. Dataset Information

```
Dataset Shape: (569, 33)

Column Names:
['id', 'diagnosis', 'radius_mean', 'texture_mean', 'perimeter_mean', 'area_mean', 'smoothness_mean', 'compactness_mean', 'concavity_mean', 'concave points_mean', 'symmetry_mean', 'fractal_dimension_mean', 'radius_se', 'text
```

	id	diagnosis	radius_mean	texture_mean	perimeter_mean	area_mean	smoothness_mean	compactness_mean	concavity_mean	concave points_mean	...	texture_worst	perimeter_worst	area_worst	smoothness_worst	compactness_worst
0	842302	M	17.99	10.38	122.80	1001.0	0.11840	0.27760	0.3001	0.14710	...	17.33	184.60	2019.0	0.1622	0.6656
1	842517	M	20.57	17.77	132.90	1326.0	0.08474	0.07864	0.0869	0.07017	...	23.41	158.80	1956.0	0.1238	0.1866
2	84300903	M	19.69	21.25	130.00	1203.0	0.10960	0.15990	0.1974	0.12790	...	25.53	152.50	1709.0	0.1444	0.4245
3	84348301	M	11.42	20.38	77.58	386.1	0.14250	0.28390	0.2414	0.10520	...	26.50	98.87	567.7	0.2098	0.8663
4	84358402	M	20.29	14.34	135.10	1297.0	0.10030	0.13280	0.1980	0.10430	...	16.67	152.20	1575.0	0.1374	0.2050

5 rows × 33 columns

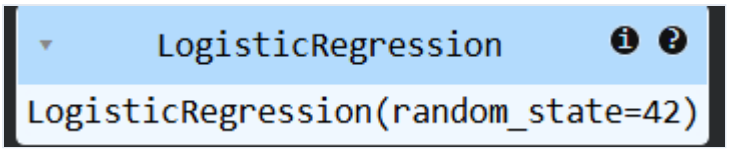
2. Data Preprocessing

```
Missing Values:
diagnosis                 0
radius_mean               0
texture_mean              0
perimeter_mean            0
area_mean                 0
smoothness_mean           0
compactness_mean          0
concavity_mean            0
concave points_mean       0
symmetry_mean             0
fractal_dimension_mean    0
radius_se                 0
texture_se                0
perimeter_se              0
area_se                   0
smoothness_se             0
compactness_se            0
concavity_se              0
concave points_se         0
symmetry_se               0
fractal_dimension_se      0
radius_worst              0
texture_worst             0
perimeter_worst           0
...
diagnosis
B    357
M    212
Name: count, dtype: int64
```

3. Train Test Split and Standardization

```
Training Features Shape: (455, 30)
Testing Features Shape: (114, 30)
Training Target Shape: (455,)
Testing Target Shape: (114,)
```

4. Logistic Regression Model Training



5. Confusion Matrix and Model Evaluation

```
Confusion Matrix:
[[71  1]
 [ 3 39]]

Precision: 0.975
Recall: 0.9285714285714286
ROC-AUC Score: 0.996031746031746
```

6. Threshold Tuning

```
Threshold: 0.3
Precision: 0.9762
Recall: 0.9762
------------------------------
Threshold: 0.4
Precision: 0.9756
Recall: 0.9524
------------------------------
Threshold: 0.5
Precision: 0.975
Recall: 0.9286
------------------------------
Threshold: 0.6
Precision: 1.0
Recall: 0.9048
------------------------------
Threshold: 0.7
Precision: 1.0
Recall: 0.9048
------------------------------
```

7. Sigmoid Function Explanation

Sigmoid Function

The sigmoid function converts the output of Logistic Regression into a probability between 0 and 1.

Its mathematical formula is:

[\sigma(z) = \frac{1}{1 + e^{-z}}]

- If the probability is 0.5 or higher, the default prediction is 1 (Malignant).
- If the probability is below 0.5, the default prediction is 0 (Benign).
- Changing the classification threshold changes how predictions are classified.

8. Sigmoid Function Graph

